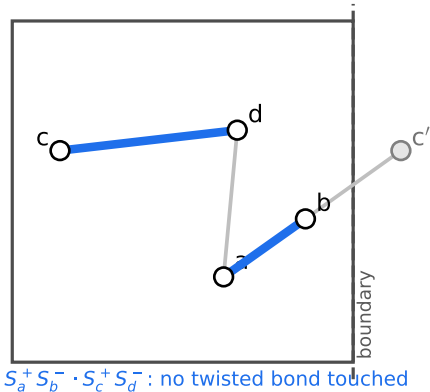


$$c_C(\varphi) = -\frac{g_4}{2} [e^{-i\mathbf{N}_A \cdot \varphi} + e^{-i\mathbf{N}_B \cdot \varphi}] = -\frac{g_4}{2} (1 + e^{-i\mathbf{w} \cdot \varphi})$$

matching A: bulk bonds only, $\mathbf{N}_A = \mathbf{0}$



matching B: uses the crossing, $\mathbf{N}_B = \mathbf{w}$

